



Habib University
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Introduction to Probability and Statistics

EE/CE 354/361 L2
Spring Semester 2025

"The most important questions of life are indeed, for the most part, really only problems of probability."

- Pierre-Simon Laplace (Théorie Analytique des Probabilités, 1812)

"In each field, we must carefully distinguish three aspects of the theory: (a) the formal logical content, (b) the intuitive background, and (c) the applications. The character, and the charm, of the whole structure cannot be appreciated without considering all three aspects in their proper relation."

- William Feller (An Introduction to Probability Theory and its Applications, 1950)

Instructor Information

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Office Hours: TBA

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Course Description

In the present world, we encounter situations where we have to make decisions on the basis of incomplete or imperfect information. The theory of probability helps provide a formal mechanism for understanding, quantifying, and dealing with uncertainty, which is ever present in our lives, pure

science, or engineering applications. Simply, by uncertainty we mean the condition when outcomes or future are not completely determined or can be captured by a deterministic function; they depend on a number of factors and perhaps just on pure chance. A lot of our present day technologies will not be possible without an understanding of how to make decisions in presence of uncertainty. These technologies include all forms of wireless communication, servers, speech processing systems, network systems and so many more. Equipping yourself with tools to deal with uncertainty will help you with whatever you wish to pursue in life.

Course Aims

This course will cover the foundations of probability, random variables and statistics, with a plethora of examples from electrical engineering, computer engineering, computer science, and everyday life. The course content can broadly be divided into three categories: -

1. Fundamentals of probability,
2. Common probability models,
3. Inferences & statistics.

We'll be working towards the following 'course objectives' in this course: -

1. Gaining an appreciation for the limitation of human intuition when faced with finding optimal solution for a problem in the presence of uncertainty.
2. Recognizing the presence of uncertainty in a scenario and modeling it using formal notions of probability.
3. Identifying the right questions to ask when drawing inferences from a probability model and interpreting results correctly.

Course Learning Outcomes (CLOs)

Course Learning outcomes		
After the completion of the course, students should be able to:		
SLO/ CLO	Description	Learning- Domain Level
CLO 1	Transform a description of random experiment into formal setting of probability theory – construct sample space, identify and specify events.	Cog - 4

CLO 2	Apply the axioms of probability and compute probabilities by counting.	Cog - 3
CLO 3	Compute probabilities of events and moments from given distributions.	Cog - 3
CLO 4	Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables;	Cog - 3
CLO 5	Analyze a random experiment for presence of dependent situations and modeling and analyzing it using notions of conditional probability.	Cog - 4

Mode of Instruction

- Two in-person lectures will be held during each week. All classes will be of 75 minutes duration.
- The LMS (Canvas) site will be used to share the syllabus, give out assignments, and share other course resources. Official course communication will take place on Canvas.
- Students are required to attend classes and submit assignments on time. Before each class, they should review the topics covered in the last class. Students' performance will be assessed using various instruments including exams, assignments, and projects.

Engagement & Participation Rules

- Students are not allowed to use their cellphones inside the classroom.
- The University's standard policies on attendance, inclusivity, office hours, and academic integrity apply in this course.

Required Texts and Materials

2
Sample Space and
Probability

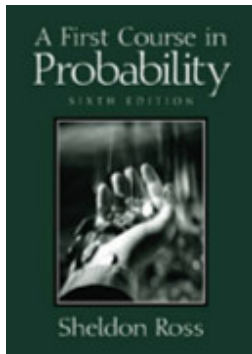
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2	100

Introduction to Probability

Authors: Dimitri P. Bertsekas, John N. Tsitsiklis

Publication Date: 2002-01-01

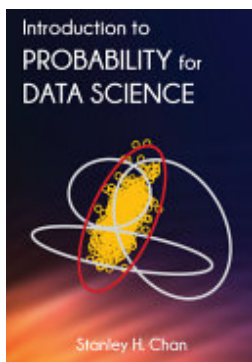
Optional Materials



A First Course in Probability

Edition: 10

Authors: Sheldon M. Ross



Introduction to Probability for Data Science

ISBN: 9781607857464

Authors: Stanley Chan

Publication Date: 2021-09-30

Assessments

No	Assessment type	Total	Weight (%)	Remarks
1	Short 15-20 minutes quiz 6 (best 5 out of 6)		20	These will mostly be based on the lectures and practice questions from the lectures.
2	Homework Assignments 3		15	One assignment each from Chapter No 1, Chapter No 2, & Chapter No 3
3	Programming Assignment 1		10	Students (in group of two) have to complete a Python-based programming assignment

4	Term Exam 1	1	15	The format will be released later.
5	Term Exam 2	1	15	The format will be released later.
6	End-term Exam	1	25	An in-person exam will be conducted that would be comprehensive

Grading Scale

Letter Grade	GPA Points	Percentage
A+	4.00	[95-100]
A	4.00	[90-95)
A-	3.67	[85-90)
B+	3.33	[80-85)
B	3.00	[75-80)
B-	2.67	[70-75)
C+	2.33	[67-70)
C	2.00	[63-67)
C-	1.67	[60-63)
F	0.00	[0, 60]

Note: [a, b) is a range of numbers from a to b where a is included in the range and b is not.

Attendance Policy

Students are expected to maintain 100% attendance in the courses at HU. However, all students must maintain class attendance per the attendance threshold document shared by the RO to deal with any unforeseen situations at your end during the semester. If one cannot participate in any session, inform the instructor within 24 hours with a reason. Noncompliance will eventually lead to withdrawing/failing the student(s) from this course. Attendance will be marked manually in the class per the University's attendance policy.

Week-Wise Schedule (Tentative)

Week No	Topics to be covered	Out	In

1	Introduction; Sets, Sample Space, Probabilistic Models, Axioms,		
2	Discrete & Continuous Models; Conditional Probability;		
3	Total Probability; Bayes' rule	Quiz 1	Quiz 1
4	Independence; Counting	HW Assignment 1	
5	Discrete Random variables, PMF; Common discrete RVs		HW Assignment 1
6	Common discrete RVs; Functions of RVs;	Term Exam I	Term Exam I
7	Mean, and Variance of known Discrete Random Variables; Joint PMF	Quiz 2 Programming Assignment 1	Quiz 2,
8	Conditional PMF; Conditional Expectation; Independence of RVs	HW Assignment 2	Programming Assignment 1
9	Continuous RVs and PDF; Expectation; CDF	Quiz 3	Quiz3 HW Assignment 2
10	Normal RVs;	Term Exam II	Term Exam II
11	Joint PDFs;	Quiz 4 HW Assignment 3	Quiz 4
12	Conditional PDFs;	Programming Assignment 2	
13	Continuous Bayes' Rule		HW Assignment 3
14	Derived distributions;	HW Assignment 4	Programming Assignment 2
15	Statistical Inferencing; Bayesian inferencing; (only if time permits)	Quiz 5	Quiz 5 HW Assignment 4
16	End Term Exam	ET	ET

Final Exam Policy

An in-person final exam will be conducted that would be comprehensive (in terms of content).

Late Submission Policy

- Quizzes, Exams: No late submission is allowed in QUIZZES and EXAMS. All the quizzes will be held during class.
- Assignments: Late submission of assignments will incur a 50% penalty.

Academic Integrity

Each student in this course is expected to abide by the Habib University Student Honor Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work.

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by Habib University regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

- a. Plagiarism: Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. As per University policy, plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated, // bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.
- b. Cheating: The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.
- c. Collusion: Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so.

All violations of academic integrity will also be immediately reported to the Student Conduct Office.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having

possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

Should copying occur, the student who copied work from another student and the student who gave material to be copied will both be in violation of the Student Code of Conduct.

If you wish to use generative-AI tools to complete any of your assessments, you must first obtain permission from your course instructor. AI generated work will not be accepted in all classes or even all assessments. The instructor's permission is required. If the permission is granted, you should declare its use and properly cite the source of the generated content. Failing to identify AI written or assisted work is academic dishonesty and will be treated as any case of plagiarism by the university.

The principle for academic integrity is that your submissions must be substantially your own work and that any work that is not originally your thought must be identified and credited. If the use of AI tools is prohibited in the course, respect the rules and do not use these tools for assessments. The fundamental purpose of assessment is to learn, synthesize information and explain new connections and interpretations that arise from your secondary research. Be aware that unauthorized use of AI tools for assessments can result in a conduct case being filed. This can have serious consequences for your academic standing and future career opportunities.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

Program Learning Outcomes (For Administrative Review)

Upon graduation, students will have the following abilities:

- PLO 1: Engineering Knowledge: An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
- PLO 2: Problem Analysis: An ability to identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

- PLO 3: Design/Development of Solutions: An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
- PLO 4: Investigation: An ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions.
- PLO 5: Modern Tool Usage: An ability to create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering activities, with an understanding of the limitations.
- PLO 6: The Engineer and Society: An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solution to complex engineering problems.
- PLO 7: Environment and Sustainability: An ability to understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate knowledge of and need for sustainable development.
- PLO 8: Ethics: Apply ethical principles and commit to professional ethics, responsibilities, and norms of engineering practice.
- PLO 9: Individual and Team Work: An ability to work effectively, as an individual or in a team, on multifaceted and /or multidisciplinary settings.
- PLO 10: Communication: An ability to communicate effectively, orally as well as in writing, on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PLO 11: Project Management: An ability to demonstrate management skills and apply engineering principles to one's own work, as a member and/or leader in a team, to manage projects in a multidisciplinary environment.
- PLO 12: Lifelong Learning: An ability to recognize importance of, and pursue lifelong learning in the broader context of innovation and technological developments.

	Program Learning Outcomes (PLOs) mapped to MATH 310/EE 354 SLOs/CLOs				
	SLOs/CLOs of the course are designed to cater following PLOs: PLO 1 : Engineering Knowledge, PLO 2 : Problem Analysis				
		Distribution of CLO weights for each PLO			
PLOs	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1	25%	25%	25%	25%	
PLO 2					100%

	CLO #01	CLO #02	CLO #03	CLO #04	CLO #05
HW Assignment 1					
HW Assignment 2					
HW Assignment 3					
Quiz No 01					
Quiz No 02					
Quiz No 03					
Quiz No 04					
Quiz No 05					
Programming Assignment					
Term Exam 1					
Term Exam 2					
End-Term Exam					

Recording Policy

Only asynchronous and synchronous online sessions will be conducted and recorded via MS Teams. Link to the recordings will be available to all students on Canvas Learning Management System.

Accommodations for Students with Disabilities

In compliance with the Habib University policy and equal access laws, I am available to discuss appropriate academic accommodations that may be required for student with disabilities. Requests

for academic accommodations are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made. Students are encouraged to register with the Office of Academic Performance to verify their eligibility for appropriate accommodations.

Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Habib University is committed to providing an atmosphere for learning that respects diversity. While working together to build this community we ask all members to:

- share their unique experiences, values and beliefs
- be open to the views of others
- honor the uniqueness of their colleagues
- appreciate the opportunity that we have to learn from each other in this community
- value each other's opinions and communicate in a respectful manner
- keep confidential discussions that the community has of a personal (or professional) nature
- use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Habib community

Office Hours Policy

Every student enrolled in this course must meet individually with the course instructor during course office hours at least once during the semester. The first meeting should happen within the first five weeks of the semester but must occur before midterms. Any student who does not meet with the instructor may face a grade reduction or other penalties at the discretion of the instructor and will have an academic hold placed by the Registrar's Office.